

p04	p05	p06	p07	p08	p09	p10	p11	p12	p13	p14	p15	p16	p17	p18	p19	p20	p21	p22	p23	p24	Ailment
410204.1015	8834.625206	4.809561543	42.03342928	92.31988178	27.00482547	35.25769311	5	18.91414333	96.54616158	61.41815302	116.9365583	5.041603178	81.51819551	49.96804444	11.82820167	26.2734386	97.39832295	0.649259333	0.01899765	2.891521804	Healthy
421416.3821	7551.743543	4.650226096	40.20991897	85.49627316	28.29254482	34.05903352	16.51621267	24.47310287	110.9539437	71.85007167	98.79127562	5.7999744	109.0892982	41.42025868	40	20.12530129	85.01860282	1.023011641	0.033543201	1.499165355	Diabetes
267971.2381	6030.085265	4.623835802	46.89354796	85.23409435	32.4087426	32.08201518	16.75228725	22.57654873	91.45038296	74.16855189	58.84493792	4.964791704	92.35793752	48.04832875	10	25.09726182	74.37811105	0.862689577	0.032801142	2.929267153	Thalassemia
204385.8555	4507.019659	4.352787524	38.89184104	93.11322561	27.77841205	34.04040759	17.56905017	19.61877827	94.85840382	63.33499386	91.43792887	4	70.4480024	56.57459577	18.28221107	27.7636204	65.04147579	0.911119962	0.026087759	1.32109599	Anemia
384908.4868	7096.110188	4.696468718	50.80361446	88.33509114	28.80012022	35.29964314	8.221810718	22.59647967	94.0993024	60	92.25150653	4.884779938	77.14708776	43.83339575	10.25959265	26.93730229	92.95219192	0.687617541	0.032591219	0.340852279	Thalassemia
253512.7979	6596.950284	4.517225227	50.87837332	99.79386236	31.22852614	33.44048334	14.13870704	22.91661197	114.0130193	71.45227037	128.6057419	4.799186587	101.3556328	43.85683296	34.54475876	13.27010583	95.34866586	0.670887494	0.015395066	2.824214961	Diabetes
150000	5552.874403	4.677963328	43.44618517	88.74125308	28.90115495	35.03177121	18.30206415	21.31987022	95.0196528	64.46224608	90.06825542	4.922088237	113.9428108	52.12831115	35.25914656	16.75157718	82.56034372	0.85643256	0.017377683	1.994477349	Thrombocytopenia
175450.6998	10534.1209	4.963555571	47.75796071	87.03800864	28.54340868	33.99578442	7.506472518	20.61104449	116.1093033	71.41743054	89.88421564	5.406764035	70	55.42341509	17.91262074	19.4596996	84.68567749	1.099387138	0.029283213	2.286053282	Thrombocytopenia
207236.9042	6797.411862	5.059889327	50.17079308	96.31524527	27.86117268	32	21.36623062	22.19075703	117.0293308	62.92809325	62.75711789	4.013630796	71.78462133	52.73148262	32.7282015	22.35392656	83.78257014	0.727195993	0.022476884	0.120797221	Thalassemia
175450.6998	10534.1209	4.963555571	47.75796071	87.03800864	28.54340868	33.99578442	7.506472518	20.61104449	116.1093033	71.41743054	89.88421564	5.406764035	70	55.42341509	17.91262074	19.4596996	84.68567749	1.099387138	0.029283213	2.286053282	Thrombocytopenia
163176.0808	4245.27034	4.666693799	45.78127401	86.15767842	32.84894861	34.95399879	9.482192985	21.07018878	113.232018	60.80781088	77.97864026	4.752202533	96.31755078	60	12.5825873	19.32208244	88.51165655	0.722940311	0.024335688	1.093620996	Anemia
197108.6559	9118.115414	4.201916231	50.88251245	81.88301442	27.83396919	33.12434898	6.944273344	21.37235713	102.740308	62.34224309	69.2479749	5.186980662	78.72163455	57.24057947	21.38036649	18.49679039	100	1.170977667	0.016726945	0.534858453	Diabetes
213466.3277	5026.591171	4.416037342	42.38082955	91.95249544	30.02667157	33.83247923	10.45446028	22.60583251	115.8882825	68.64860519	55.88045124	4.36938193	88.85903696	51.01406635	31.59405857	12.64483518	99.36923205	0.875927198	0.01168339	2.664668557	Anemia
334889.5985	10426.85901	4.922610476	51.5559859	95.43641684	28.66283538	35.95466039	9.08157362	20.08610389	113.283786	64.38231796	69.64778292	5.15235634	115.0948361	59.74346442	15.20028622	21.92344447	86.23813362	1.086776772	0.019970355	0.336996375	Diabetes
320517.3523	8819.422571	4.904059088	44.67845914	81.09251608	31.49260308	33.29162096	19.27937603	19.50434823	101.076174	69.46596936	121.1775933	5.34898367	109.4203125	45.63480154	18.14686224	21.97460245	81.98708121	0.902740226	0.011183625	2.279024865	Healthy
333945.6844	4852.51291	4.452190305	38.34306909	85.07402296	27.34848184	35.27696845	16.79101646	18.63052813	110.016779	74.47995094	50	5.54671494	72.66468038	44.45227885	11.89571543	31.31795164	95.30892547	1.053106322	0.024159105	2.736168209	Anemia
406897.5413	9378.93592	4.802065077	38.68808253	97.26522412	28.7986127	32.00394855	14.8106776	20.77931922	102.381738	65.82442547	79.83325191	5.104167554	70.70417534	56.5431316	22.73926988	28.54938751	84.13834559	0.790386922	0.031478472	0.784324361	Anemia
239489.7568	9232.441784	4.981950176	48.9854851	82.88571623	28.53314806	32.63996397	12.2810351	24.75272445	92.47508835	74.96107668	68.52290186	4.369471174	129.6106367	46.34804088	35.33685525	40	81.8135175	0.867976282	0	0.723767139	Diabetes
316933.815	9902.87768	4.746836001	42.1183798	94.24309655	32.70620288	32.87275484	13.068665	18.85761978	97.00585132	63.18239313	81.34102283	4.136213372	97.07607109	45.32780289	31.79685208	31.44646058	68.35462971	1.06264827	0.012060048	0.098086329	Thalassemia
208347.8264	10064.42903	4.417182782	44.90264095	92.50189039	28.62776393	33.53263064	18.38050383	24.9	120	75.65293896	121.8051321	4.963892119	73.96517399	40	19.98904876	14.27829658	95.20196493	0.668686967	0.04	2.836754643	Diabetes
188831.6446	9074.685188	4.392146886	47.81154898	96.32873627	32.8050233	33.74123712	12.28583214	22.4472442	117.8050194	76.14391794	82.28206759	4.022466895	104.9946331	53.65789064	13.88780947	28.43711004	83.88838291	0.790230312	0.011821951	1.953563933	Anemia
207236.9042	6797.411862	5.059889327	50.17079308	96.31524527	27.86117268	32	21.36623062	22.19075703	117.0293308	62.92809325	62.75711789	4.013630796	71.78462133	52.73148262	32.7282015	22.35392656	83.78257014	0.727195893	0.022476884	0.120797221	Thalassemia
200540.9048	9057.625798	5.181170445	46.75478309	87.37571823	31.74994296	34.99888204	9.26197248	19.25864552	94.87878468	70.09214139	124.9798534	5.01344127	74.3947782	57.29707458	29.30423962	30.30888812	99.19842175	0.953971471	0.019265793	2.333377718	Thrombocytopenia
373368.1805	9224.732107	4.662813961	49.71850825	81.96606536	31.70896547	33.98729239	20.84226104	22.28827865	108.3975869	70.40565718	69.13999458	5.583370926	105.9899114	47.19704047	18.56813027	23.16867836	78.31607349	1.018860729	0.003360794	2.823153036	Healthy
316933.815	9902.87768	4.746836001	42.1183798	94.24309655	32.70620288	32.87275484	13.068665	18.85761978	97.00585132	63.18239313	81.34102283	4.136213372	97.07607109	45.32780289	31.79685208	31.44646058	68.35462971	1.06264827	0.012060048	0.098086329	Thalassemia
320517.3523	8819.422571	4.904059088	44.67845914	81.09251608	31.49260308	33.29162096	19.27937603	19.50434823	101.076174	69.46596936	121.1775933	5.34898367	109.4203125	45.63480154	18.14686224	21.97460245	81.98708121	0.902740226	0.011183625	2.279024865	Healthy
213466.3277	5026.591171	4.416037342	42.38082955	91.95249544	30.02667157	33.83247923	10.45446028	22.60583251	115.8882825	68.64860519	55.88045124	4.36938193	88.85903696	51.01406635	31.59405857	12.64483518	99.36923205	0.875927198	0.01168339	2.664668557	Anemia
328408.5634	8348.500737	4.28264618	41.22449432	92.89346595	30.55972097	33.47573998	20.41939585	20.08578887	103.5309581	72.40009183	73.32330826	4.222698128	70.58568794	54.70726405	27.02688733	29.01073751	96.44552973	0.924198192	0.003960418	1.559186641	Anemia
316933.815	9902.87768	4.746836001	42.1183798	94.24309655	32.70620288	32.87275484	13.068665	18.85761978	97.00585132	63.18239313	81.34102283	4.136213372	97.07607109	45.32780289	31.79685208	31.44646058	68.35462971	1.06264827	0.012060048	0.098086329	Thalassemia
197108.6559	9118.115414	4.201916231	50.88251245	81.88301442	27.83396919	33.12434898	6.944273344	21.37235713	102.740308	62.34224309	69.2479749	5.186980662	78.72163455	57.24057947	21.38036649	18.49679039	100	1.170977667	0.016726945	0.534858453	Diabetes
386435.1501	6166.368571	4.540243442	42.60158672	91.99517178	31.88848592	35.85191946	18.8025155	23.31730921	90	76.27222647	85.4326983	5.676115367	97.42834504	58.42622478	15.63737194	33.08880825	87.79404534	1.115219553	0.019195476	3	Thalassemia
333945.6844	4852.51291	4.452190305	38.34306909	85.07402296	27.34848184	35.27696845	16.79101646	18.63052813	110.016779	74.47995094	50	5.54671494	72.66468038	44.45227885	11.89571543	31.31795164	95.30892547	1.053106322	0.024159105	2.736168209	Anemia
206007.5926	6578.62699	4.340579627	44.46469586	80	28.52148045	34.46402552	6.785893983	20.18247069	107.8695258	73.09583787	110.7058641	5.252940768	115.6437018	47.71733197	23.35486097	32.04178747	84.5750845	0.779134701	0.014688258	2.370218688	Anemia
320517.3523	8819.422571	4.904059088	44.67845914	81.09251608	31.49260308	33.29162096	19.27937603	19.50434823	101.076174	69.46596936	121.1775933	5.34898367	109.4203125	45.63480154	18.14686224	21.97460245	81.98708121	0.902740226	0.011183625	2.279024865	Healthy
175450.6998	10534.1209	4.963555571	47.75796071	87.03800864	28.54340868	33.99578442	7.506472518	20.61104449	116.1093033	71.41743054	89.88421564	5.406764035	70	55.42341509	17.91262074	19.4596996	84.68567749	1.099387138	0.029283213	2.286053282	Thrombocytopenia
436387.9914	6876.066904	4.561847399	50.65589198	99.92997255	28.28478141	33.799583	16.38006846	19.48633885	113.9061066	70.50106304	66.50019243	5.792967381	86.4560317	42.32831208	36.38409959	16.82748752	61.1353117	1.113209465	0.031689361	2.0648233	

- How to 'understand this data set?'
 - What kind of general information will help?
 - What kind of 'statistics' will help?
 - What kind of 'visualization' will help?

RangeIndex: 2371 entries, 0 to 2370			
Data columns (total 25 columns):			
#	Column	Non-Null Count	Dtype
0	p01	2371 non-null	float64
1	p02	2371 non-null	float64
2	p03	2370 non-null	float64
3	p04	2370 non-null	float64
4	p05	2370 non-null	float64
5	p06	2370 non-null	float64
6	p07	2370 non-null	float64
7	p08	2370 non-null	float64
8	p09	2370 non-null	float64
9	p10	2370 non-null	float64
10	p11	2370 non-null	float64
11	p12	2370 non-null	float64
12	p13	2370 non-null	float64
13	p14	2370 non-null	float64
14	p15	2369 non-null	float64
15	p16	2369 non-null	float64
16	p17	2370 non-null	float64
17	p18	2370 non-null	float64
18	p19	2370 non-null	float64
19	p20	2370 non-null	float64
20	p21	2370 non-null	float64
21	p22	2370 non-null	float64
22	p23	2371 non-null	float64
23	p24	2371 non-null	float64
24	Ailment	2369 non-null	object

p01	0
p02	0
p03	1
p04	1
p05	1
p06	1
p07	1
p08	1
p09	1
p10	1
p11	1
p12	1
p13	1
p14	1
p15	2
p16	2
p17	1
p18	1
p19	1
p20	1
p21	1
p22	1
p23	0
p24	0
Ailment	2

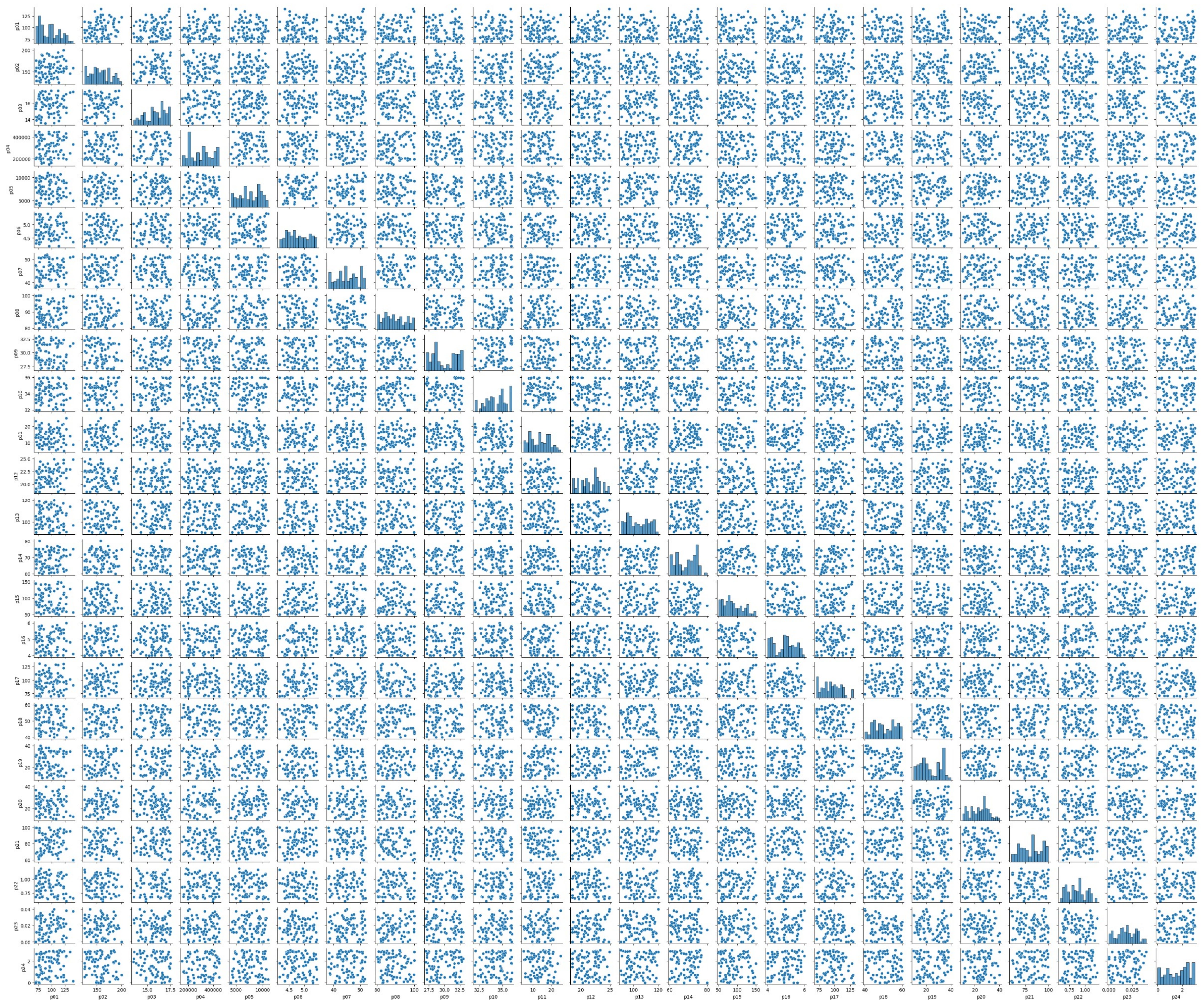
```
--- Missing Value Diagnostics ---
Number of rows with missing values: 4
Row Index | Count of Missing Values
-----
Index 223 | 2 missing value(s)
Index 762 | 1 missing value(s)
Index 769 | 1 missing value(s)
Index 773 | 20 missing value(s)
-----
Total duplicate rows found (exact matches): 2282
-----
```


	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
p01	2371.0	95.723590	18.391307	70.0	79.045276	94.910515	111.766345	140.00	0.413277	-0.869748
p02	2371.0	157.056068	20.086634	125.0	140.428465	157.334147	172.881629	200.00	0.252003	-0.904394
p03	2370.0	15.875821	1.108117	13.5	14.841100	15.976076	16.716185	17.50	NaN	NaN
p04	2370.0	299642.033487	92072.554351	150000.0	207236.904200	308502.921600	375652.950400	450000.00	NaN	NaN
p05	2370.0	7576.392817	1975.407145	4000.0	5779.743222	7692.153435	9232.441784	11000.00	NaN	NaN
p06	2370.0	4.781409	0.334514	4.2	4.475088	4.731108	5.078057	5.40	NaN	NaN
p07	2370.0	45.176374	4.130930	38.0	42.006265	44.982337	48.754763	52.00	NaN	NaN
p08	2370.0	89.368838	5.810423	80.0	85.074023	88.564827	94.243097	100.00	NaN	NaN
p09	2370.0	30.012760	1.964813	27.0	28.292545	29.618746	31.846506	33.00	NaN	NaN
p10	2370.0	34.288155	1.128564	32.0	33.440483	34.464026	35.032851	36.00	NaN	NaN
p11	2370.0	13.853883	5.197321	5.0	8.966785	13.806624	18.302064	25.00	NaN	NaN
p12	2370.0	21.549882	1.757576	18.5	20.182471	21.608173	22.752112	24.90	NaN	NaN
p13	2370.0	103.698003	8.497371	90.0	96.340427	102.868580	110.953944	120.00	NaN	NaN
p14	2370.0	68.951467	5.348610	60.0	63.725242	70.092141	74.160955	80.00	NaN	NaN
p15	2369.0	88.251874	26.647529	50.0	68.522902	82.282068	110.705664	150.00	NaN	NaN
p16	2369.0	4.904990	0.565004	4.0	4.369382	4.963892	5.362492	6.00	NaN	NaN
p17	2370.0	94.604228	15.925514	70.0	81.656821	93.982213	106.078371	130.00	NaN	NaN
p18	2370.0	50.659141	5.686841	40.0	45.463188	49.968044	55.577814	60.00	NaN	NaN
p19	2370.0	23.748406	8.590197	10.0	16.539442	21.740315	32.551651	40.00	NaN	NaN
p20	2370.0	23.445505	7.405850	10.0	16.927927	24.467557	28.437110	40.00	NaN	NaN
p21	2370.0	81.200410	11.370365	60.0	70.244560	82.500163	91.104974	100.00	NaN	NaN
p22	2370.0	0.868532	0.152437	0.6	0.727196	0.862670	0.988298	1.20	NaN	NaN
p23	2371.0	0.018510	0.010415	0.0	0.011663	0.017378	0.027957	0.04	0.051306	-0.830708
p24	2371.0	1.610787	0.919862	0.0	0.723767	1.803444	2.370219	3.00	-0.228260	-1.275177

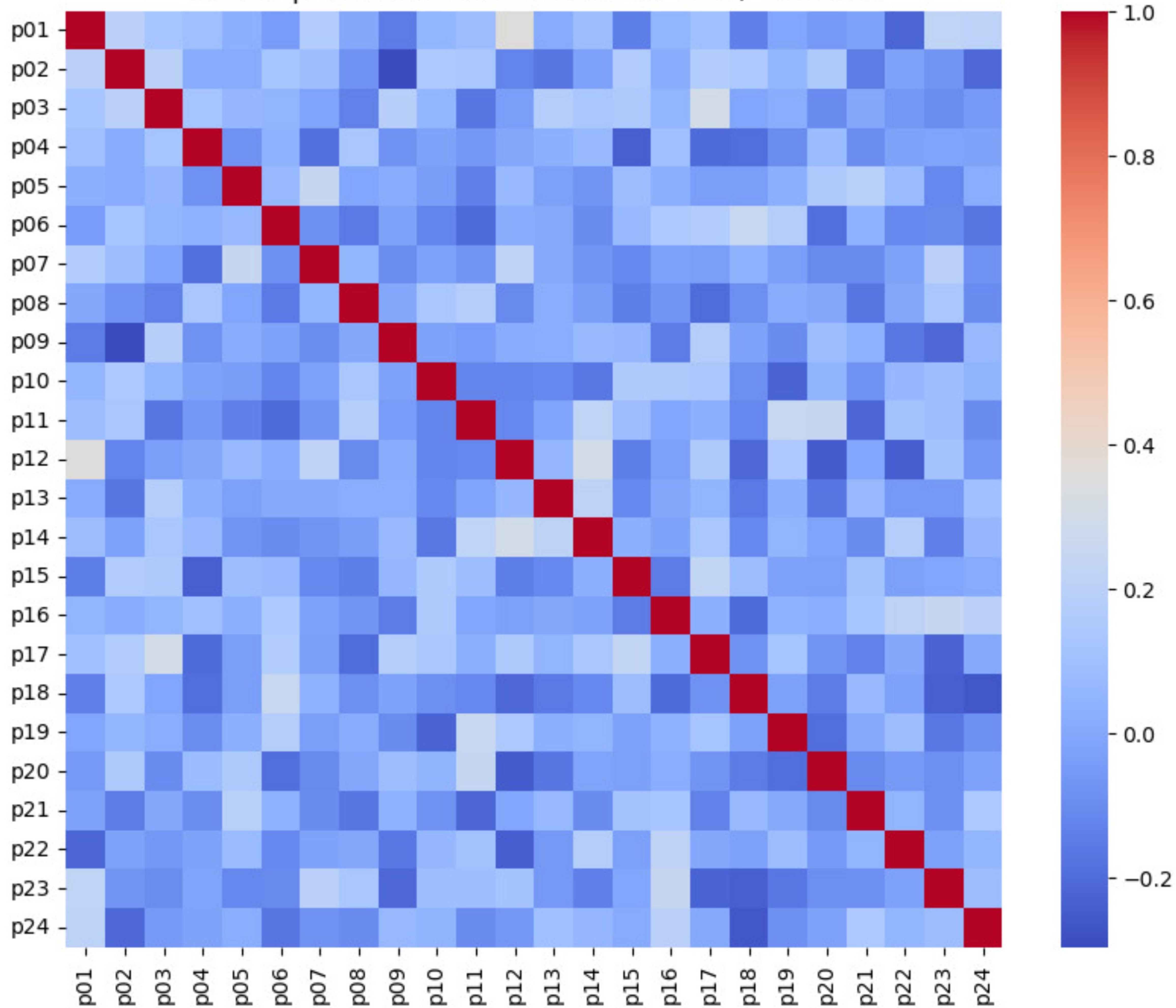
- ① Scale issues exist ↗
- ② Due to missing values
Some values cannot be calculated
- ③ Rows with missing values need to be repaired / dropped.
- ④ If decision = repair, then with what do we replace the missing values?
How do we decide?

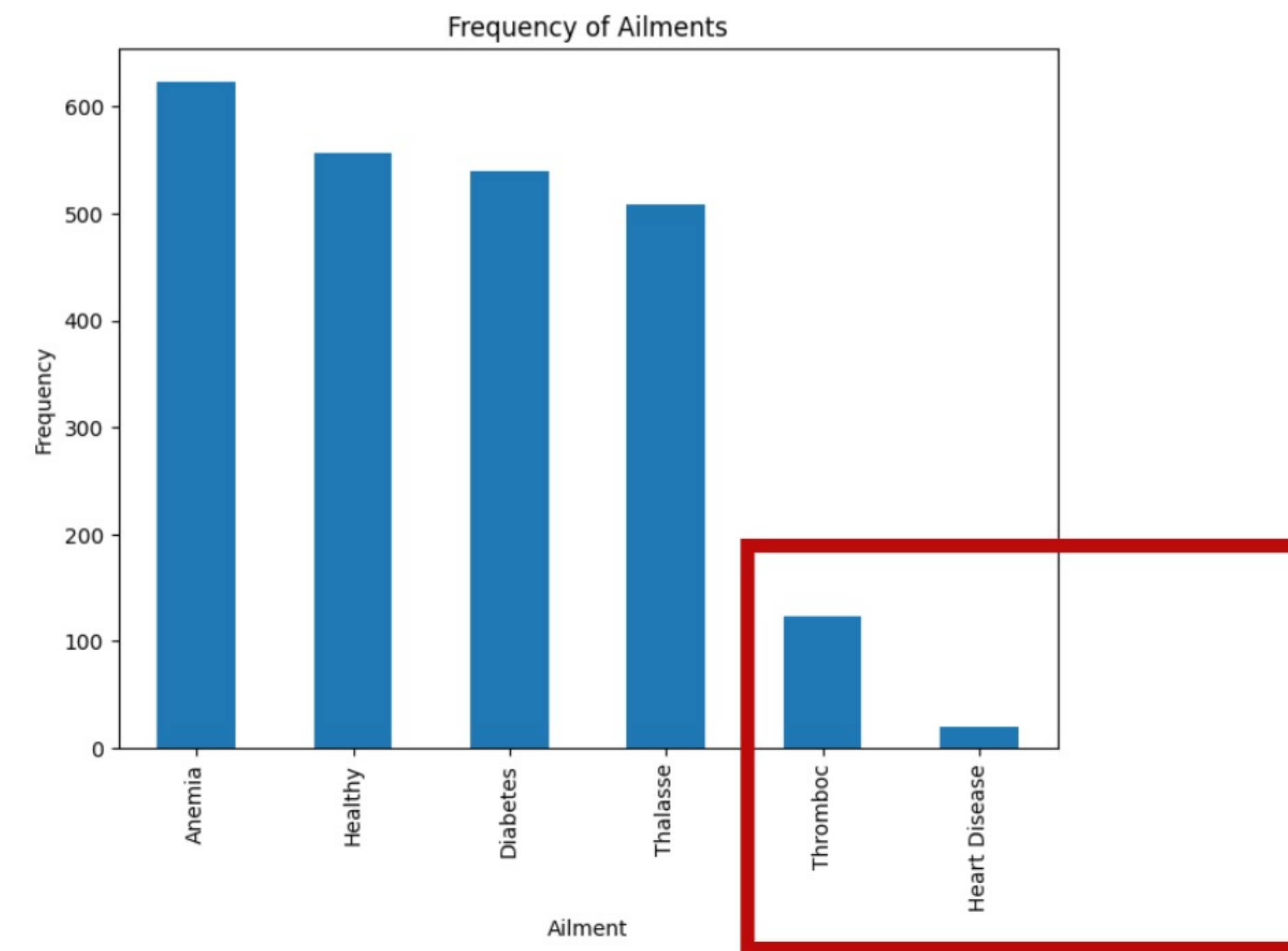
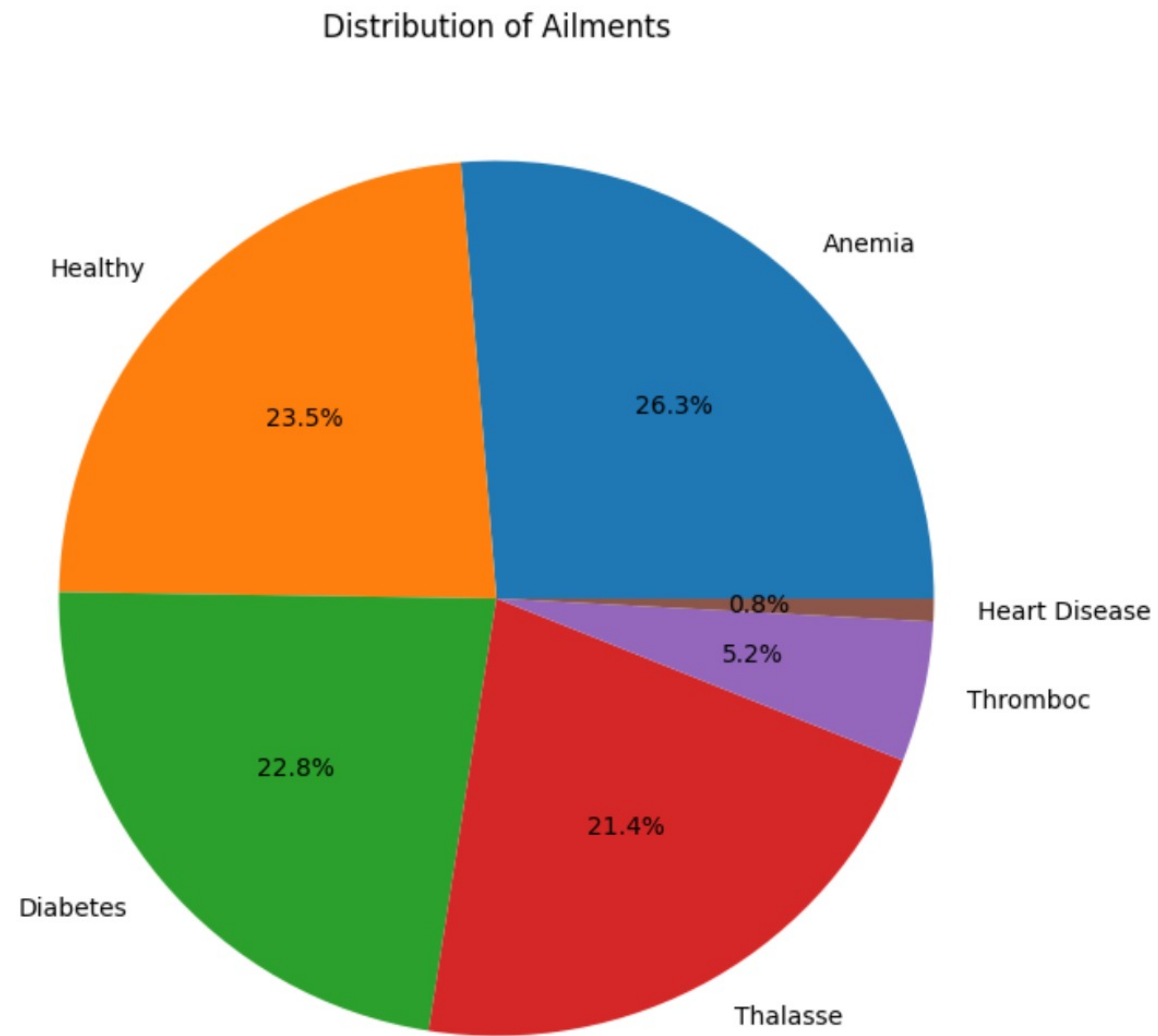
DESCRIPTIVE STATS

Pairwise scatter plot of all columns, except 'Ailments' (columns DO NOT seem to be correlated)

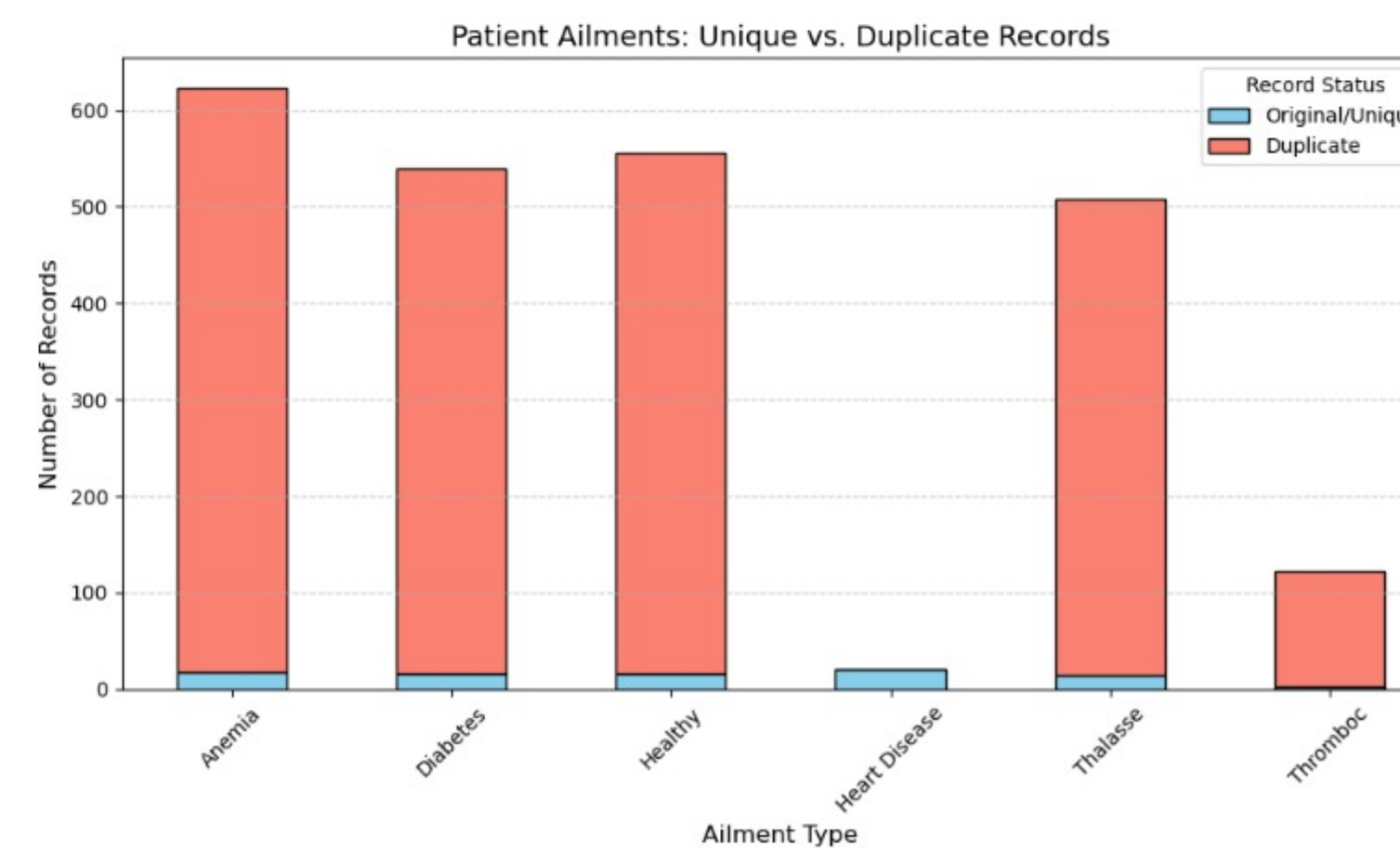


Heatmap of Correlation Coefficients - Train / Test Data

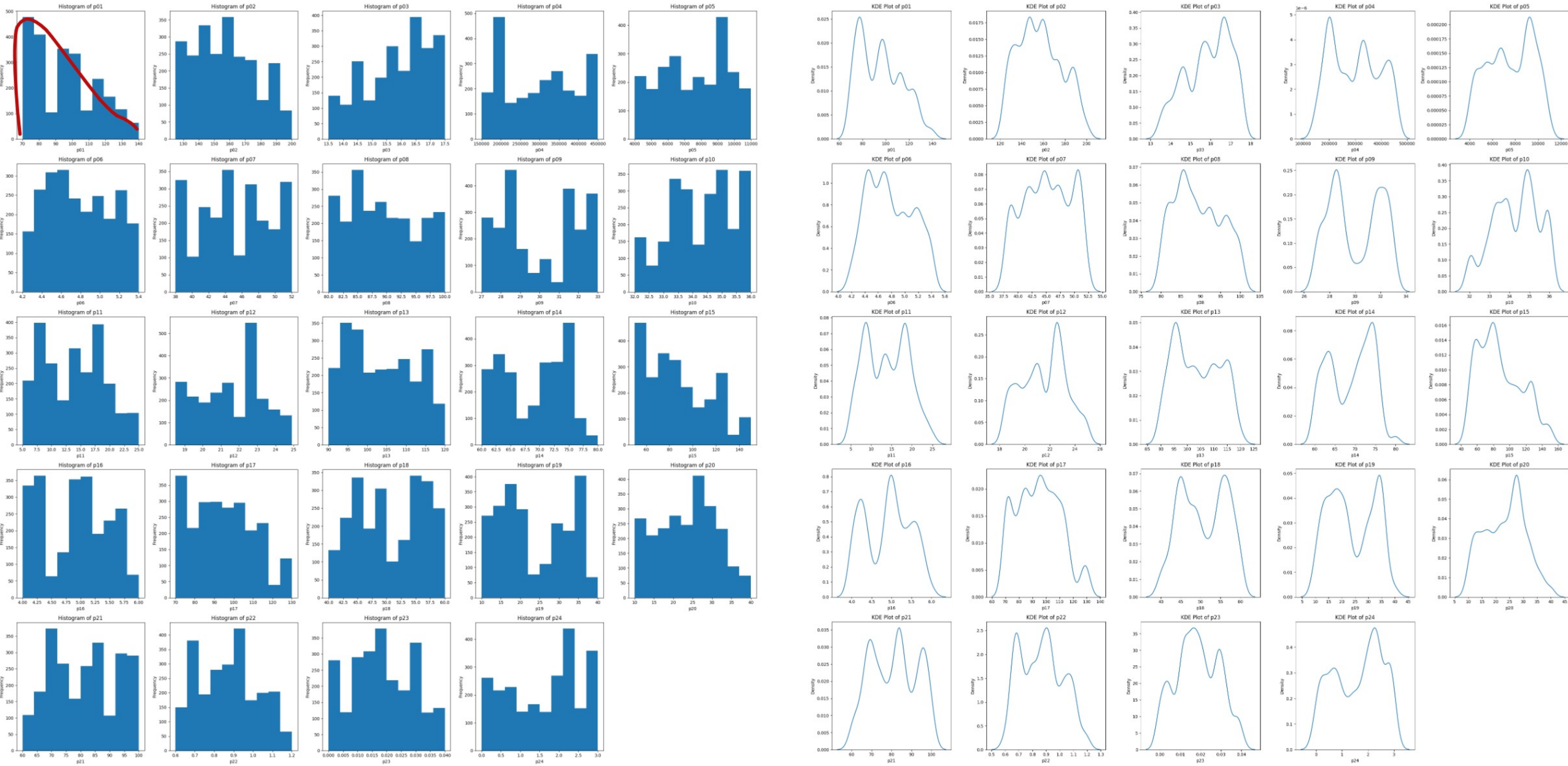




Case of
data imbalance

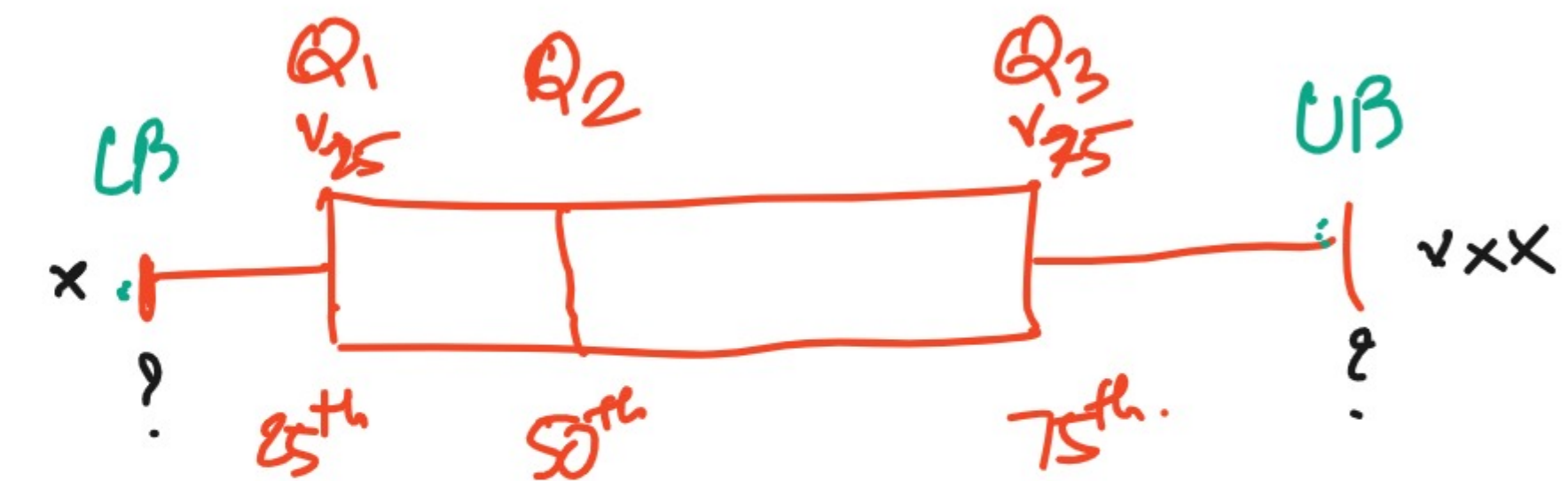


Histograms and Kernel Density Estimates: Convey data distributions within columns



Box Plots :

- They are based on percentile distribution of data
- They are created by sorting numeric data in ascending order and then identifying the 25th, 50th and 75 percentiles
- The value corresponding to the 50th percentile is nothing but the **median**
- The **Inter-quartile Range (IQR)** is defined as the range of values between the 25th and 75th percentile
- IQR is used to identify **outliers**
- Values to the left of LB / right of UB are typically considered to be **outliers**
- Note: This definition of outliers is 'mathematical'! It is important to exercise judgement based on **domain** considerations.

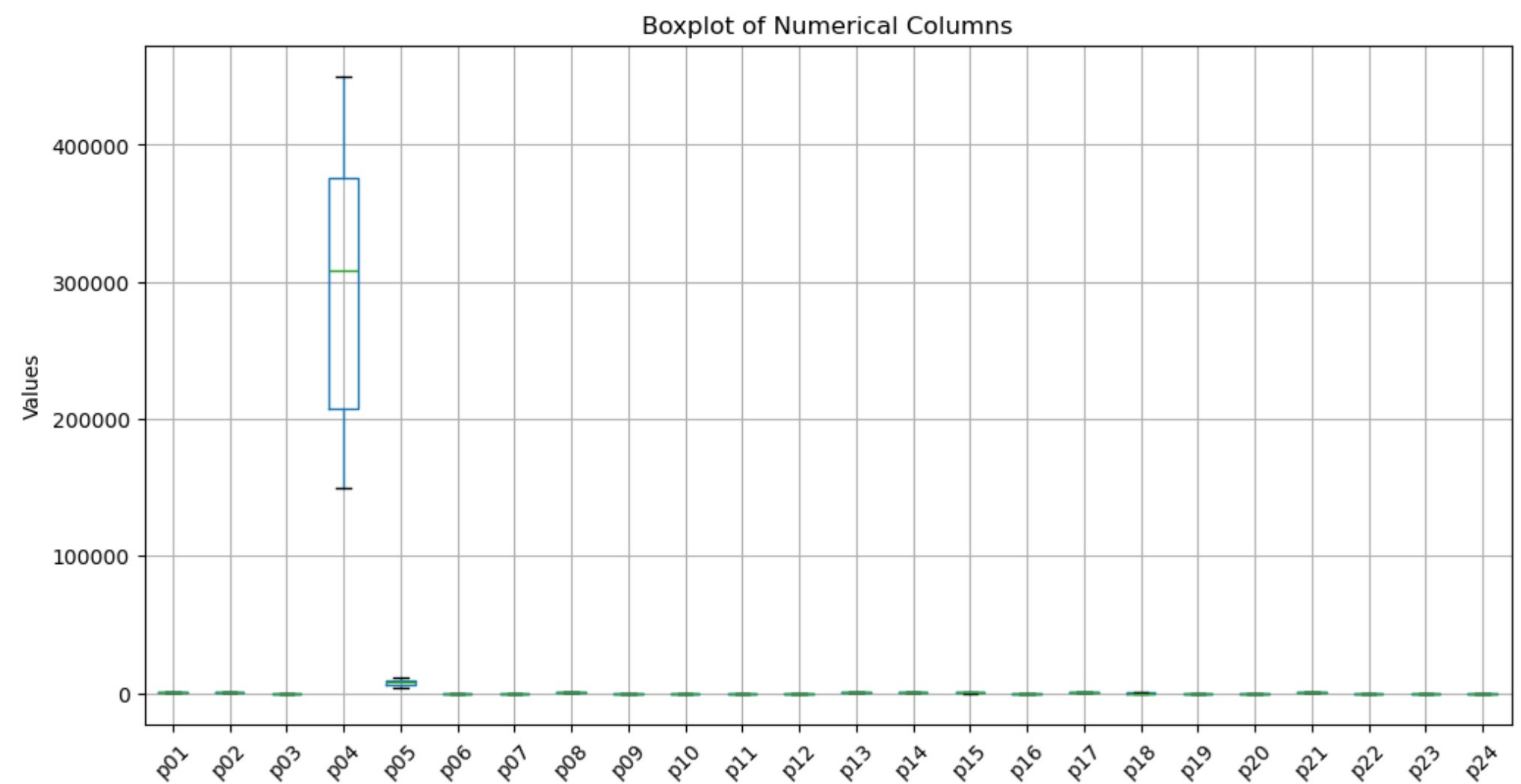
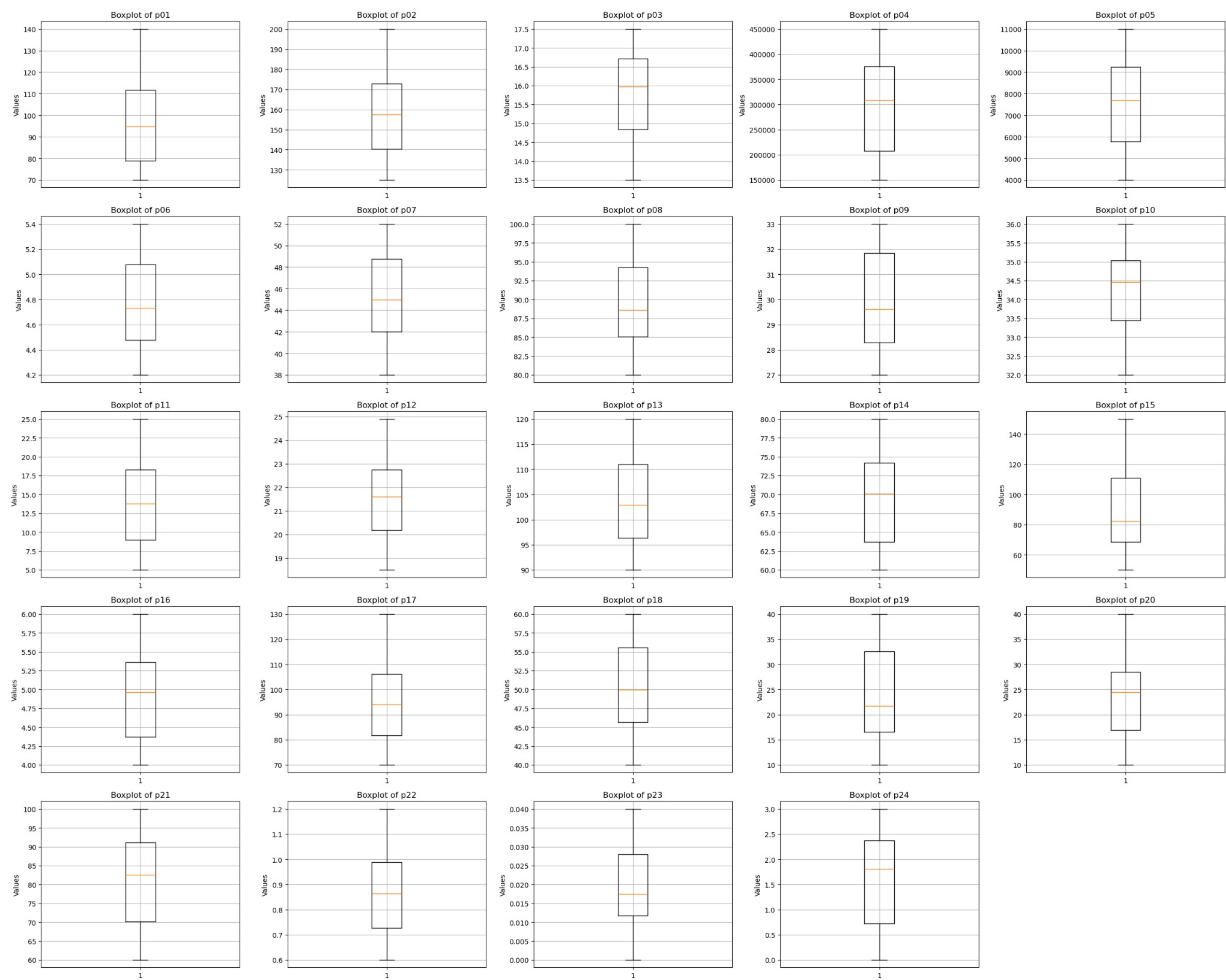


$$\begin{aligned} \text{I.Q.R} &= Q_3 - Q_1 \\ &= (V_{75}) - (V_{25}) \end{aligned}$$

$$LB = Q_1 - 1.5 \times \text{IQR}$$

$$HB = Q_3 + 1.5 \times \text{IQR}$$

- The following plot shows Box-Plots for individual columns of the blood-test dataset.
- The lower figure shows all the column box-plots plotted to a common scale. It is very clear from the lower figure that one of the column is way off in scale - a potential problem situation in the context of being used in creating ML models. This situation needs to be fixed using data scaling operations.
- It is also clear (surprisingly) that there are **no outliers** in any of the columns!



All box plots drawn to a common scale. Dominance of one of the columns is highlighted.